HKUST — MATH 2111 Matrix Algebra and Applications
Fall 2026–2027 Midterm Test

Sample B Solutions
Total: 100 marks

Qn. 1 (40 marks)

(1) **(B).** If $h \neq 2$, the last row is $0 = h - 2$, so the system is inconsistent. If $h = 2$, there are two pivot variables and one free variable, hence infinitely many solutions.

(2) **(C).** Since $v_4 = 2v_1 - v_2$, deleting v_4 does not change the span. Also

$$v_1 = \tfrac{1}{2}(v_4 + v_2), \qquad v_2 = 2v_1 - v_4,$$

so deleting v_1 or v_2 also leaves the span unchanged. Because $\{v_1, v_2, v_3\}$ is linearly independent, deleting v_3 changes the span.

(3) **(C).** A pivot in every row makes T onto $\mathbb{R}^3$. A 3×5 matrix cannot have a pivot in all five columns, so T is not one-to-one.

(4) **(D).** For AX to exist, X must have 3 rows; for XB to exist, X must have 4 columns. Thus X is 3×4.

(5) **(A).** The row interchange contributes -1, the row scaling contributes -2, and the row replacement contributes 1. Hence

$$\det B = (-1)(-2)(-3) = -6.$$

(6) **(D).** The determinant of the matrix having the three vectors as columns is

$$\det \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & k \end{bmatrix} = k - 2.$$

Dependence occurs exactly when $k = 2$.

(7) **(C).** Areas are multiplied by $|\det A| = 3$, so the image area is $3 \cdot 5 = 15$.

(8) **(C).** Right-multiply by $(A^T)^{-1}$:

$$X = B(A^T)^{-1} = B(A^{-1})^T.$$

(9) **(A).** A nonzero solution of $Ax = 0$ means that the columns are linearly dependent. For square A, this is equivalent to singularity, so $\det A = 0$.

(10) **(B).** Reduced row-echelon form is unique. Ordinary echelon form need not be unique; equal row counts do not imply row equivalence; and row operations need not preserve the actual column space.

qn.	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ans.	B	C	C	D	A	D	C	C	A	B

Qn. 2 (14 marks) The augmented matrix is

$$\left[\begin{array}{cccc|c} 1 & 1 & -1 & 2 & 2 \\ 2 & -1 & 1 & 1 & 1 \\ 4 & 1 & -1 & 5 & t+3 \\ 3 & -3 & 3 & 0 & 2-t \end{array}\right].$$

Apply

$$R_3 \leftarrow R_3 - 2R_1, \quad R_3 \leftarrow R_3 - R_2, \qquad R_4 \leftarrow R_4 + R_1, \quad R_4 \leftarrow R_4 - 2R_2.$$

Then

$$\left[\begin{array}{cccc|c} 1 & 1 & -1 & 2 & 2 \\ 2 & -1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & t-2 \\ 0 & 0 & 0 & 0 & 2-t \end{array}\right].$$

(a) Consistency requires both last entries to vanish, so $\boxed{t = 2}$.

(b) At $t = 2$, row reduction gives

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Let $x_3 = s$ and $x_4 = r$. Then $x_1 = 1 - r$ and $x_2 = 1 + s - r$, hence

$$\boxed{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} + r \begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \end{bmatrix}, \quad s, r \in \mathbb{R}.}$$

(c) The coefficient matrix has rank $2 < 4$. Its columns are linearly dependent, and $x \mapsto Ax$ is not onto $\mathbb{R}^4$.

Qn. 3 (16 marks) (a)

$$AB = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ -1 & 1 & 1 \end{bmatrix} = I_3.$$

Thus

$$\boxed{A^{-1} = B = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ -1 & 1 & 1 \end{bmatrix}.}$$

Qn. 3 (continued)

(b) From $A^TXA = D$, multiply on the left by $(A^T)^{-1}$ and on the right by A^{-1}:

$$X = (A^T)^{-1}DA^{-1} = (A^{-1})^T DA^{-1} = B^T DB.$$

First,

$$DB = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 3 & 3 \\ 2 & 3 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ -1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 2 & 0 \\ 2 & 1 & 0 \end{bmatrix}.$$

Therefore

$$X = B^T(DB) = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ 1 & 2 & 0 \\ 2 & 1 & 0 \end{bmatrix} = \boxed{\begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & -1 \\ 1 & -1 & 0 \end{bmatrix}}.$$

A direct substitution gives $A^TXA = D$.

Qn. 4 (15 marks) (a) Write $x = c_1u_1 + c_2u_2 + c_3u_3$. Comparing coordinates gives

$$c_1 = x_2 - x_3, \qquad c_2 = x_2 - x_1, \qquad c_3 = x_1 - x_2 + x_3.$$

Hence

$$T(x) = c_1 \begin{bmatrix} 3 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_3 \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \boxed{\begin{bmatrix} x_1 + 2x_2 - x_3 \\ x_2 + x_3 \end{bmatrix}},$$

$$\boxed{[T] = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \end{bmatrix}}.$$

(b) Solve

$$x_1 + 2x_2 - x_3 = 2, \qquad x_2 + x_3 = 1.$$

Let $x_3 = s$. Then $x_2 = 1 - s$ and $x_1 = 3s$, so

$$\boxed{x = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}, \quad s \in \mathbb{R}.}$$

The nonzero kernel vector $[3, -1, 1]^T$ shows that T is not one-to-one. The standard matrix has a pivot in each row, so T is onto $\mathbb{R}^2$.

Qn. 5 (15 marks) The columns of W are the indicated coordinate combinations of the columns of V, so

$$W = VC(t), \qquad C(t) = \begin{bmatrix} 1 & 0 & 0 & 1 \\ t & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & t & 1 \end{bmatrix}.$$

(a) Expanding along the first row,

$$\det C(t) = \det \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & t & 1 \end{bmatrix} - \det \begin{bmatrix} t & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & t \end{bmatrix} = 1 - t^2.$$

Therefore

$$\boxed{\det W = \det V \det C(t) = 3(1 - t^2).}$$

(b) Linear dependence occurs exactly when $\det W = 0$, hence

$$\boxed{t = 1 \quad \text{or} \quad t = -1.}$$

(c) For $t = 1$,

$$\begin{aligned} -w_1 + w_2 - w_3 + w_4 &= -(v_1 + v_2) + (v_2 + v_3) \\ &\quad - (v_3 + v_4) + (v_4 + v_1) = 0. \end{aligned}$$

Thus

$$\boxed{-w_1 + w_2 - w_3 + w_4 = 0}.$$

For $t = -1$,

$$\begin{aligned} -w_1 - w_2 + w_3 + w_4 &= -(v_1 - v_2) - (v_2 + v_3) \\ &\quad + (v_3 - v_4) + (v_4 + v_1) = 0. \end{aligned}$$

Thus

$$\boxed{-w_1 - w_2 + w_3 + w_4 = 0}.$$

End of solutions